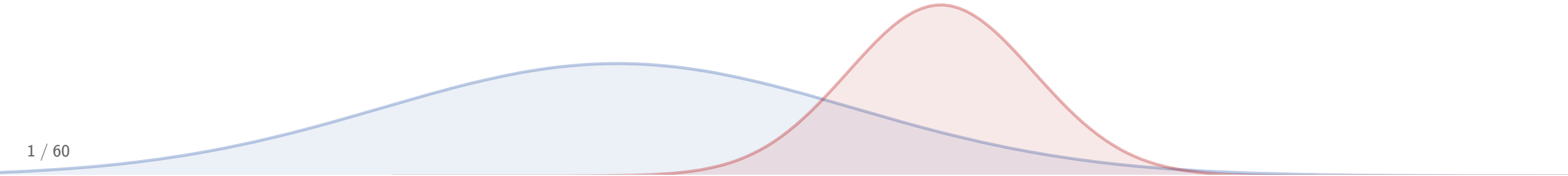
Linear Classification

Part 3: How can we make accurate predictions based on data, and how accurate can they be?

Probabilistic Machine Learning

University of Virginia
Farzad Farnoud

October 15, 2026



Rain tomorrow, and the next word

Problem setup: classification

Input: a feature vector $\mathbf{x} \in \mathbb{R}^d$.

Label: $y \in \mathcal{Y}$, a finite set of classes.

Training data: labelled examples

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}.$$

Goal: learn to predict the label of a new input $\mathbf{x}$.

RAIN TOMORROW

$\mathbf{x}$: humidity, pressure

$y \in \{0, 1\}$: dry or rainy

THE NEXT WORD

$\mathbf{x}$: features of the text so far

y: next token, one of K



AI-generated illustration

A probabilistic approach

A purely predictive classifier outputs only a label $\hat{y}(\mathbf{x})$. A probabilistic classifier estimates $p(y \mid \mathbf{x})$ for each label, then predicts the most probable one.

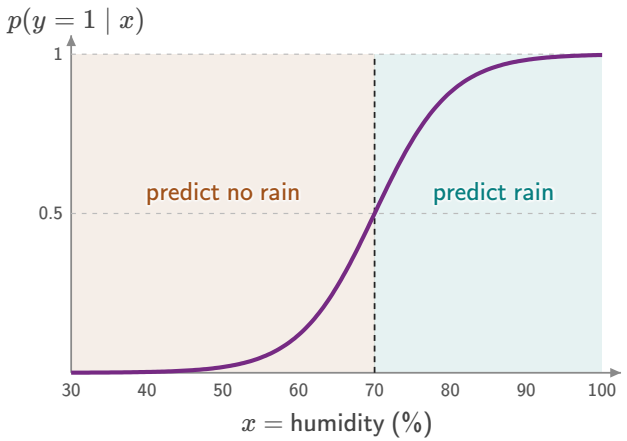
- Probabilities say how sure the prediction is.
- Smooth dependence on the parameters lets gradient descent fit the model.
- Sampling generates data, such as the text an LLM writes.

Suppose we know $p(y \mid \mathbf{x})$. Predict the most probable class:

$$\hat{y}(\mathbf{x}) = \arg\max_j p(y = j \mid \mathbf{x}).$$

This minimizes the probability of misclassification,

$$\mathcal{L} = P\big(\hat{y}(\mathbf{X}) \neq Y\big).$$



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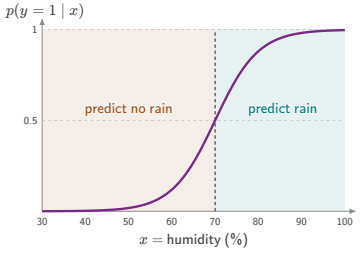
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Derivation

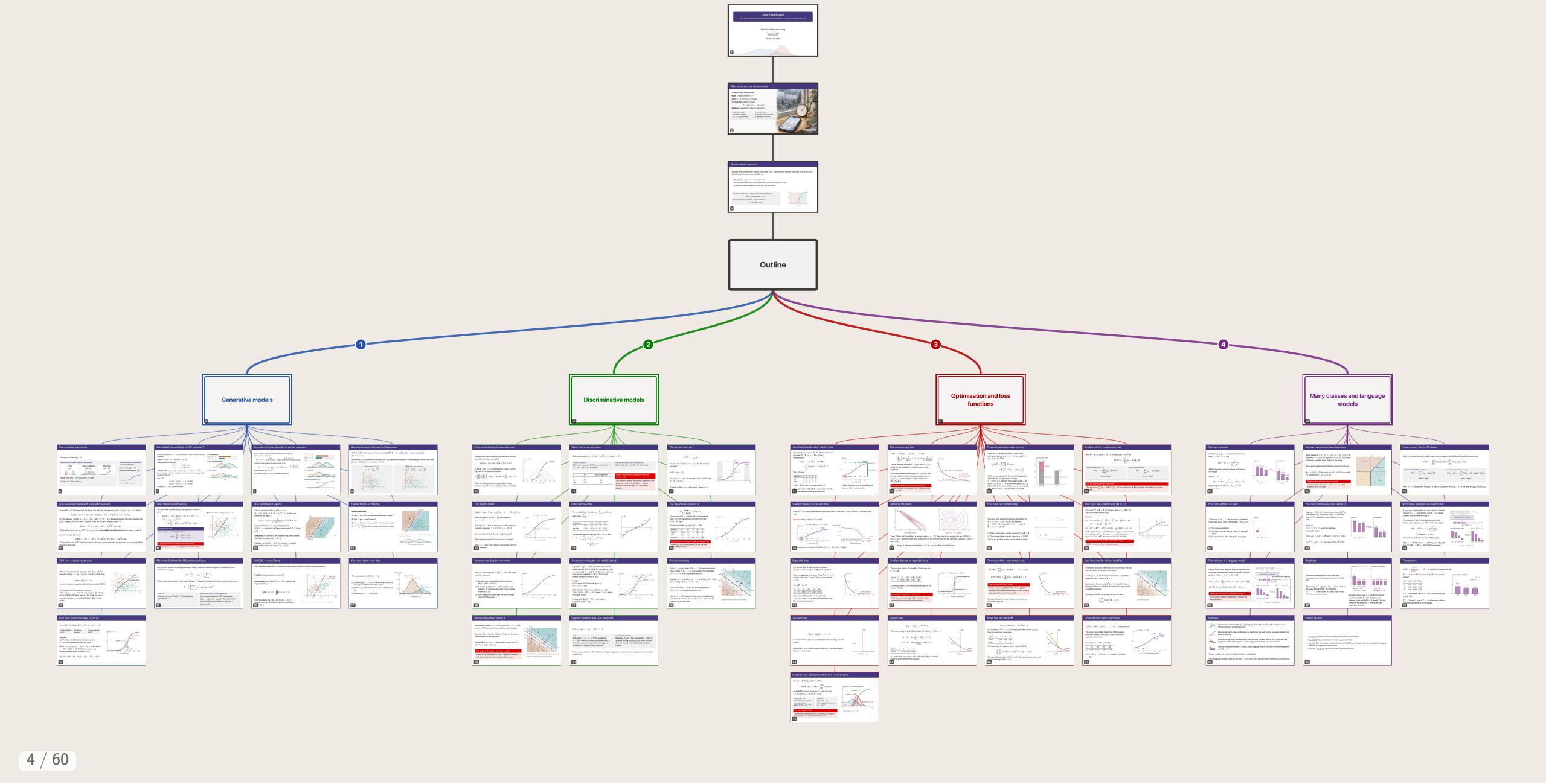
Given $\mathbf{X} = \mathbf{x}$, the prediction $\hat{y}(\mathbf{x})$ is wrong with probability $1 - p\big(y = \hat{y}(\mathbf{x}) \mid \mathbf{x}\big)$.

This is smallest when $p\big(y = \hat{y}(\mathbf{x}) \mid \mathbf{x}\big)$ is largest, that is, for the most probable class.

Averaging over $\mathbf{X}$,

$$P\big(\hat{y}(\mathbf{X}) \neq Y\big) = \mathbb{E}\Big[1 - p\big(y = \hat{y}(\mathbf{X}) \mid \mathbf{X}\big)\Big],$$

which is smallest when every term is.


Linear Classification
Probabilistic Machine Learning
Chapter 10
Andrew Jin, 2025
Rain tomorrow, and the next word
A probabilistic approach
Outline
1
Generative models
2
Discriminative models
3
Optimization and loss functions
4
Many classes and language models

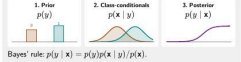
Generative models

5

Two modeling approaches

Two ways to obtain $p(y \mid \mathbf{x})$:

Generative: model how the data arise



Bayes' rule: $p(y \mid \mathbf{x}) = p(y)p(\mathbf{x} \mid y)/p(\mathbf{x})$.

Examples: LDA, QDA, naive Bayes.

Discriminative: model the posterior directly

Parametrize $p(y \mid \mathbf{x})$ without modeling $p(\mathbf{x} \mid y)$.



Example: logistic regression.

6

Which label is more likely at 70% humidity?

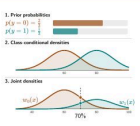
Example (illustrative): $y = 1$ for rain tomorrow; x is the afternoon humidity in percent.

Priors: $\pi = p(y = 1) = \frac{1}{4}$ and $p(y = 0) = \frac{3}{4}$.

Class-conditional densities:

$$X \mid Y = 1 \sim \mathcal{N}(80, 10^2),$$
$$X \mid Y = 0 \sim \mathcal{N}(60, 10^2).$$

Joint densities: $w_1(x) = p(y = 1)p(x \mid y = 1) = p(x, y = 1)$. By Bayes' rule $p(y = j \mid x) = w_j(x)/p(x)$, so the class with the larger $w_j(x)$ is the more likely one.



At $x = 70$:

$$w_1(70) = \tfrac{1}{4}\varphi(70 \mid y = 1) = 0.0081,$$
$$w_0(70) = \tfrac{3}{4}\varphi(70 \mid y = 0) = 0.0161.$$

Since $w_0(70) > w_1(70)$, predict **no rain**.

7

Normalize the joint densities to get the posterior

For any humidity x, the joint densities of the previous slide add up to $p(x) = w_1(x) + w_0(x)$, so

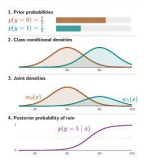
$$p(y = 1 \mid x) = \frac{w_1(x)}{w_1(x) + w_0(x)} = \frac{\pi\, p(x \mid y = 1)}{\pi\, p(x \mid y = 1) + (1 - \pi) p(x \mid y = 0)}$$

Dividing the numerator and the denominator by $w_1(x)$,

$$p(y = 1 \mid x) = \frac{1}{1 + \frac{w_0(x)}{w_1(x)}} = \frac{1}{1 + e^{-\ln(w_1(x)/w_0(x))}} = \sigma\left(\ln \frac{w_1(x)}{w_0(x)}\right),$$

with the sigmoid $\sigma(u) = \frac{1}{1+e^{-u}}$.

As x varies, $\sigma\left(\ln(w_1(x)/w_0(x))\right)$ traces the curve in step 4.

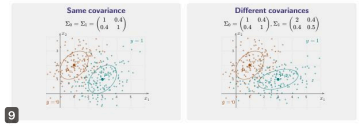


8

Gaussian class-conditionals in d dimensions

With $\mathbf{x} \in \mathbb{R}^d$, each class has a Gaussian density $\mathbf{X} \mid Y = j \sim \mathcal{N}(\boldsymbol{\mu}_j, \Sigma_j)$ and prior probability $p(y = j) = \pi_j$.

Illustration: x_1 is standardized humidity and x_2 is standardized pressure. Rain has higher humidity and lower pressure. Compare two covariance choices:



9

Exponential-family class-conditionals

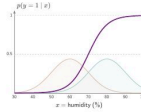
Suppose each class-conditional has sufficient statistic $\mathbf{x}$ and the same base term $f(\mathbf{x})$:

$$p(\mathbf{x} \mid y = j) = \exp\left(\boldsymbol{\eta}_j^T \mathbf{x} + f(\mathbf{x}) + g_j\right).$$

In Bayes' rule, $f(\mathbf{x})$ cancels between classes, and the log ratio of the posteriors is linear:

$$\ln \frac{p(y = 1 \mid \mathbf{x})}{p(y = 0 \mid \mathbf{x})} = (\boldsymbol{\eta}_1 - \boldsymbol{\eta}_0)^T \mathbf{x} + \ln \frac{\pi_1}{\pi_0} + g_1 - g_0.$$

Thus the binary posterior is a sigmoid of a linear function of $\mathbf{x}$. Why not model that posterior directly?



20

LDA: Gaussian classes with a shared covariance

Suppose $\Sigma_j = \Sigma$ for every class. By Bayes' rule, with the joint density $w_j(\mathbf{x}) = \pi_j\, p(\mathbf{x} \mid y = j)$ of slide 7,

$$\ln p(y = j \mid \mathbf{x}) - \ln p(y = 0 \mid \mathbf{x}) = \ln w_j(\mathbf{x}) - \ln p(\mathbf{x}) = \ln \pi_j + \ln p(\mathbf{x} \mid y = j) - \ln p(\mathbf{x}).$$

For the Gaussian, $\ln p(\mathbf{x} \mid y = j) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_j)^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}_j)$ plus a normalizing term that depends only on Σ. Dropping that term and $-\ln p(\mathbf{x})$, which are the same for every class ($\doteq$),

$$\ln p(y = j \mid \mathbf{x}) \doteq \ln \pi_j - \tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_j)^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}_j).$$

The quadratic form $(\mathbf{x} - \boldsymbol{\mu}_j)^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}_j)$ is the **squared Mahalanobis distance** from $\mathbf{x}$ to $\boldsymbol{\mu}_j$ under Σ.

Expand the quadratic form:

$$\ln p(y = j \mid \mathbf{x}) \doteq -\tfrac{1}{2}\mathbf{x}^T \Sigma^{-1} \mathbf{x} + \boldsymbol{\mu}_j^T \Sigma^{-1} \mathbf{x} + \ln \pi_j - \tfrac{1}{2} \boldsymbol{\mu}_j^T \Sigma^{-1} \boldsymbol{\mu}_j.$$

The quadratic term $\mathbf{x}^T \Sigma^{-1} \mathbf{x}$ is the same for every class and cancels when log-posteriors are compared. What remains is linear in $\mathbf{x}$.

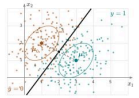
10

LDA: the decision boundary

For two classes, subtracting the log-posteriors of slide 10 gives

$$\ln p(y = 1 \mid \mathbf{x}) - \ln p(y = 0 \mid \mathbf{x}) = \boldsymbol{\beta}^T \mathbf{x} + \gamma,$$
$$\boldsymbol{\beta}^T = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \Sigma^{-1},$$
$$\gamma = \ln \frac{\pi_1}{\pi_0} - \tfrac{1}{2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \Sigma^{-1} (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_0).$$

Figure: $\Sigma_{11} = \Sigma_{00} = 1$ and $\Sigma_{12} = \Sigma_{21} = 0.4$.



Classification rule

Compare $p(y = 1 \mid \mathbf{x})$ and $p(y = 0 \mid \mathbf{x})$:

$$\hat{y}(\mathbf{x}) = \begin{cases} 1, & \boldsymbol{\beta}^T \mathbf{x} + \gamma > 0, \\ 0, & \boldsymbol{\beta}^T \mathbf{x} + \gamma < 0. \end{cases}$$

A shared covariance makes the boundary linear

It places the tie on $\boldsymbol{\beta}^T \mathbf{x} + \gamma = 0$, a hyperplane (a line in the plane).

11

LDA's posterior is logistic

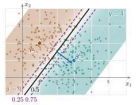
The log-posteriors differ by $\boldsymbol{\beta}^T \mathbf{x} + \gamma$, so $p(y = 0 \mid \mathbf{x})/p(y = 1 \mid \mathbf{x}) = e^{-(\boldsymbol{\beta}^T \mathbf{x} + \gamma)}$, and the two posteriors add up to 1:

$$p(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\boldsymbol{\beta}^T \mathbf{x} + \gamma)}} = \sigma(\boldsymbol{\beta}^T \mathbf{x} + \gamma).$$

Each probability level is a parallel hyperplane $\boldsymbol{\beta}^T \mathbf{x} + \gamma =$ constant; the figure marks levels 0.25, 0.5 and 0.75.

Direction: $\boldsymbol{\beta}$ is normal to the boundary and points toward the fastest increase in $p(y = 1 \mid \mathbf{x})$.

Position: for fixed $\boldsymbol{\beta}$, γ shifts the boundary. Its signed distance from the origin along $\boldsymbol{\beta}$ is $-\gamma/\|\boldsymbol{\beta}\|$.

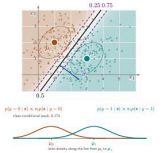


12

Explore the LDA posterior

Explore and explain.

1. Vary π_1: how do the joint densities and boundary change?
2. Drag $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$: what changes?
3. At $\pi_1 = 0.5$, then 0.6, vary s and ρ in the shared covariance $\Sigma = s^2 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$. How do the boundary and posterior change?



13

The logistic model

$$h_{\boldsymbol{\theta}}(\mathbf{x}) = p(y = 1 \mid \mathbf{x}) = \sigma(\boldsymbol{\beta}^T \mathbf{x} + \gamma), \qquad \boldsymbol{\theta} = (\boldsymbol{\beta}, \gamma).$$

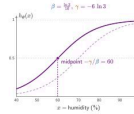
With one input x, $h_{\boldsymbol{\theta}}(x) = \frac{1}{2}$ at the midpoint $x = -\gamma/\beta$.

Example: $y = 1$ for rain tomorrow; x is the afternoon humidity in percent; $\beta = \ln 3/10$, $\gamma = -7 \ln 3$.

Question: Double both β and γ: what changes?

The midpoint stays at 70, and the curve is steeper.

With $\gamma = -6 \ln 3$ the midpoint moves to 60, with the steepness.



23

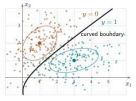
QDA, one covariance per class

With one Σ_j per class the quadratic term stays, and the boundary $\ln p(y = 1 \mid \mathbf{x}) = \ln p(y = 0 \mid \mathbf{x})$ is of the form

$$\mathbf{x}^T A \mathbf{x} + \boldsymbol{\beta}^T \mathbf{x} + \gamma = 0,$$

a curve in the plane: quadratic discriminant analysis (QDA).

Alternatively, add the quadratic features $\boldsymbol{\phi}(\mathbf{x}) = (x_1, \ldots, x_d,\ x_i x_j \text{ for } 1 \le i \le j \le d)$. A linear rule in $\boldsymbol{\phi}(\mathbf{x})$ has the same form as LDA's rule and gives a quadratic boundary in $\mathbf{x}$, without fitting a QDA density model.



14

Maximum likelihood for LDA and naive Bayes

Let n_j be the number of training examples in class j. Maximum likelihood gives the class fraction and within-class averages:

$$\hat{\pi}_j = \frac{n_j}{n}, \qquad \hat{\boldsymbol{\mu}}_j = \frac{1}{n_j} \sum_{i:\, y_i = j} \mathbf{x}_i.$$

Center observations at their class means. Chapter 4's Gaussian result gives the shared covariance estimate:

$$\hat{\Sigma} = \frac{1}{n} \sum_{j=0}^{1} \sum_{i:\, y_i = j} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_j)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_j)^T.$$

FULL LDA

Use every entry of $\hat{\Sigma}$: $d(d+1)/2$ covariance parameters.

NAIVE BAYES WITH SHARED VARIANCES

Keep only the diagonal of $\hat{\Sigma}$. Equivalently, $\hat{D}_{kk} = \frac{1}{n} \sum_i (x_{ik} - \hat{\mu}_{y_i k})^2$: the pooled within-class sample variance of feature k (MLE; d parameters).

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Full LDA vs naive Bayes

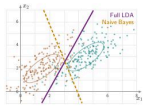
Both Gaussian classes share a covariance. Naive Bayes ignores correlations between features.

Full LDA: fit the pooled covariance $\hat{\Sigma}$.

Naive Bayes: use $\hat{D}$ with $\hat{D}_{kk} = (\hat{\Sigma})_{kk}$ and zero off-diagonal entries, so

$$p(\mathbf{x} \mid y = j) = \prod_{k=1}^{d} p(x_k \mid y = j).$$

Both boundaries are lines. LDA fits $d(d+1)/2$ covariance entries; this shared-variance naive Bayes fits d.



Both fits use the same class means and the same training points.

16

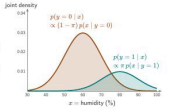
Your turn: fewer rainy days

The densities of slide 7, with $\pi = \frac{1}{4}$.

(a) What is $p(y = 1 \mid x)$ at 70% humidity, where the two class-conditional densities cross?

(b) Which way does the boundary move, and where is it?

(c) What is $p(y = 1 \mid \mathbf{x})$ at 80%?



17

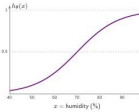
Your turn: reading the rain model

The rain model: log-odds $= \frac{\ln 2}{10}(x - 70)$, with x the humidity in percent.

(a) By what factor do the odds of rain grow from 40% to 100% humidity?

(b) A second model has $\beta = \ln 3/5$ and the same midpoint. At what humidity does it give rain the probability 0.25?

(c) What probability of rain does the second model give at 80% humidity?



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Your turn: fewer rainy days (a) to (c)

Same class densities as slide 7, with rain prior $\pi = \frac{1}{4}$.

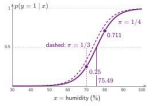
(a) Rain probability at 70%	(b) Decision boundary	(c) Rain probability at 80%

Solution

(a) The class-conditional densities are equal at $x = 70$, so the posterior equals the prior: $\frac{1}{4}$.

(b) $\ln\left(w_1(x)/w_0(x)\right) = 0.2x - 14 - \ln 3$ is zero at $x = 70 + 5 \ln 3 \approx 75.49$: the boundary moves toward the rainy mean, to about 75.5%.

(c) $\sigma(0.2 \cdot 80 - 14 - \ln 3) = \sigma(2 - \ln 3) = 0.711$.



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Decision boundary: scaling $\boldsymbol{\theta}$

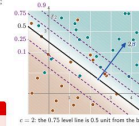
The example of slide 28: $\boldsymbol{\beta} = \ln 3\,(0.6, 0.8)$, $\gamma = -2 \ln 3$; the $h = 0.75$ level line lies one unit from the boundary.

Question: As on slide 23, doubling $\boldsymbol{\theta}$ keeps the boundary. What happens to the level lines?

Replace $\boldsymbol{\theta}$ by $c\boldsymbol{\theta}$; at $c = 2$ the boundary stays; the 0.75 level line is half a unit away.

The length of $\boldsymbol{\beta}$ sets how sharp the model is

Scaling $\boldsymbol{\theta}$ by c multiplies $\|\boldsymbol{\beta}\|$ by c, keeps the boundary, and divides each level line's distance from it by c.



$c = 2$: the 0.75 level line is 0.5 unit from the b

Illustrative samples, labels drawn from the model.

29

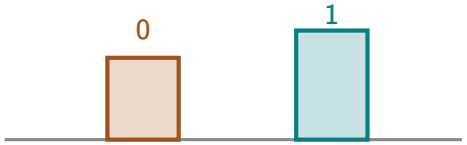
Two modeling approaches

Two ways to obtain $p(y \mid \mathbf{x})$:

Generative: model how the data arise

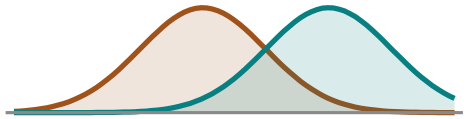
1. Prior

$p(y)$



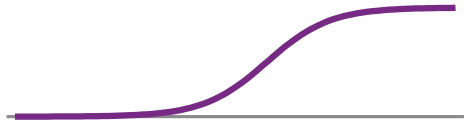
2. Class-conditionals

$p(\mathbf{x} \mid y)$



3. Posterior

$p(y \mid \mathbf{x})$

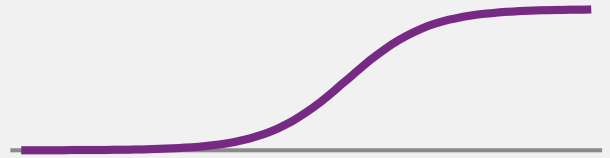


Bayes' rule: $p(y \mid \mathbf{x}) = p(y)p(\mathbf{x} \mid y)/p(\mathbf{x})$.

Examples: LDA, QDA, naive Bayes.

Discriminative: model the posterior directly

Parametrize $p(y \mid \mathbf{x})$ without modeling $p(\mathbf{x} \mid y)$.



Example: logistic regression.

Which label is more likely at 70% humidity?

Example (illustrative): $y = 1$ for rain tomorrow; x is the afternoon humidity in percent.

Priors: $\pi = p(y = 1) = \frac{1}{3}$ and $p(y = 0) = \frac{2}{3}$.

Class-conditional densities:

$$X \mid Y = 1 \sim \mathcal{N}(80, 10^2),$$
$$X \mid Y = 0 \sim \mathcal{N}(60, 10^2).$$

Joint densities: $w_j(x) = p(y = j)\, p(x \mid y = j) = p(x, y = j)$. By Bayes' rule $p(y = j \mid x) = w_j(x)/p(x)$, so the class with the larger $w_j(x)$ is the more likely one.

At $x = 70$:

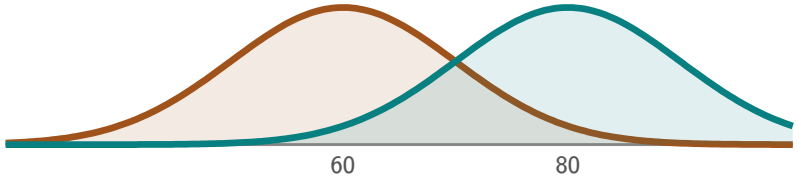
$$w_1(70) = \tfrac{1}{3}\, p(70 \mid y = 1) = 0.0081,$$
$$w_0(70) = \tfrac{2}{3}\, p(70 \mid y = 0) = 0.0161.$$

Since $w_0(70) > w_1(70)$, predict **no rain**.

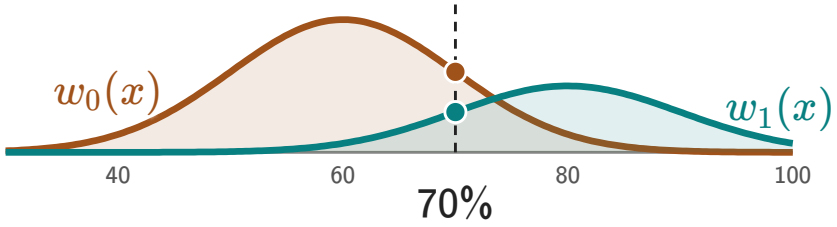
1. Prior probabilities



2. Class-conditional densities



3. Joint densities



Normalize the joint densities to get the posterior

For any humidity x, the joint densities of the previous slide add up to $p(x) = w_1(x) + w_0(x)$, so

$$p(y = 1 \mid x) = \frac{w_1(x)}{w_1(x) + w_0(x)} = \frac{\pi\, p(x \mid y = 1)}{\pi\, p(x \mid y = 1) + (1 - \pi)\, p(x \mid y = 0)}.$$

Dividing the numerator and the denominator by $w_1(x)$,

$$p(y = 1 \mid x) = \frac{1}{1 + \frac{w_0(x)}{w_1(x)}} = \frac{1}{1 + e^{-\ln(w_1(x)/w_0(x))}} = \sigma\Big(\ln \frac{w_1(x)}{w_0(x)}\Big),$$

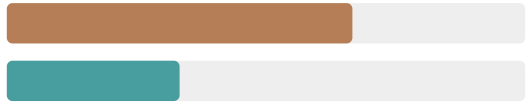
with the sigmoid $\sigma(u) = \frac{1}{1+e^{-u}}$.

As x varies, $\sigma\big(\ln(w_1(x)/w_0(x))\big)$ traces the curve in step 4.

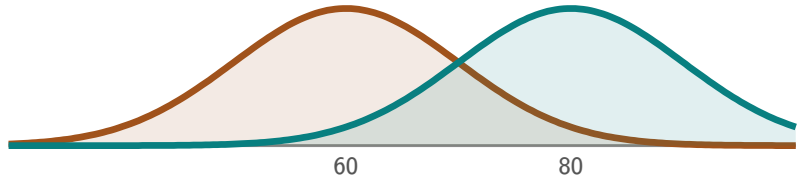
1. Prior probabilities

$p(y = 0) = \frac{2}{3}$

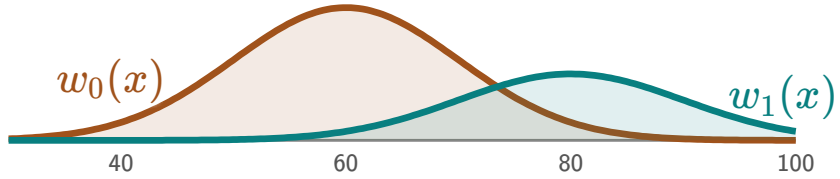
$p(y = 1) = \frac{1}{3}$



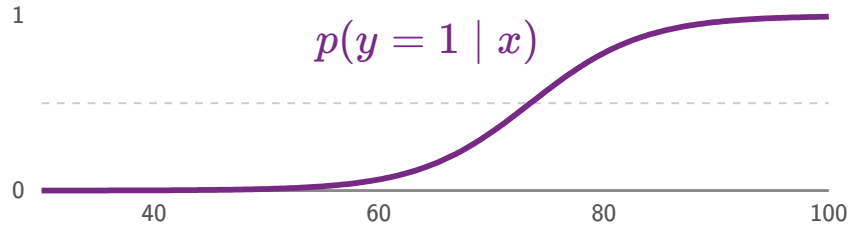
2. Class-conditional densities



3. Joint densities



4. Posterior probability of rain



Normalize the joint densities to get the posterior

For any humidity x, the joint densities of the previous slide add up to $p(x) = w_1(x) + w_0(x)$, so

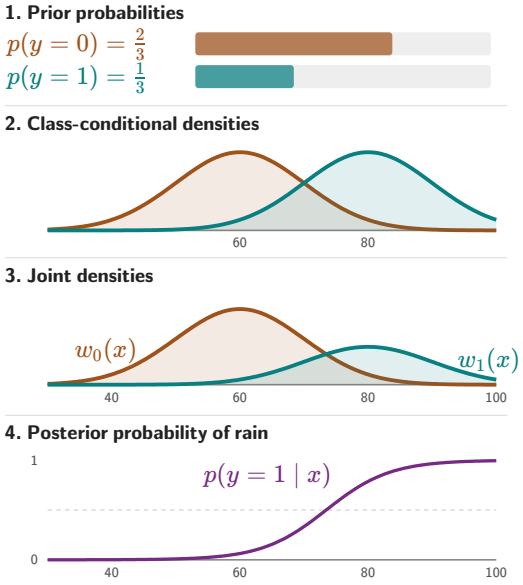
$$p(y = 1 \mid x) = \frac{w_1(x)}{w_1(x) + w_0(x)} = \frac{\pi\, p(x \mid y = 1)}{\pi\, p(x \mid y = 1) + (1 - \pi)\, p(x \mid y = 0)}.$$

Dividing the numerator and the denominator by $w_1(x)$,

$$p(y = 1 \mid x) = \frac{1}{1 + \frac{w_0(x)}{w_1(x)}} = \frac{1}{1 + e^{-\ln(w_1(x)/w_0(x))}} = \sigma\Big(\ln \frac{w_1(x)}{w_0(x)}\Big),$$

with the sigmoid $\sigma(u) = \frac{1}{1+e^{-u}}$.

As x varies, $\sigma\big(\ln(w_1(x)/w_0(x))\big)$ traces the curve in step 4.



Derivation

For the densities of the previous slide, $\pi = \frac{1}{3}$ and the Gaussian constants are equal, so

$$\begin{aligned}
\ln \frac{w_1(x)}{w_0(x)} &= -\ln 2 + \ln p(x \mid y = 1) - \ln p(x \mid y = 0) \\
&= -\ln 2 - \tfrac{1}{200}(x - 80)^2 + \tfrac{1}{200}(x - 60)^2 \\
&= \frac{x - 70}{5} - \ln 2.
\end{aligned}$$

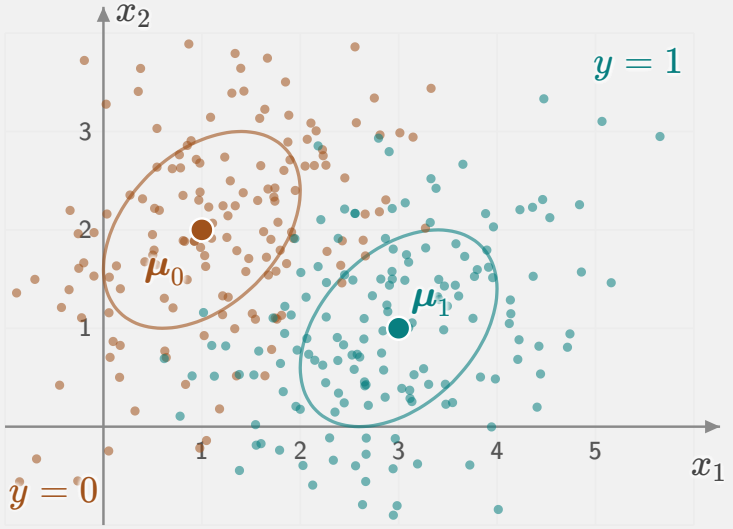
Gaussian class-conditionals in d dimensions

With $\mathbf{x} \in \mathbb{R}^d$, each class has a Gaussian density $\mathbf{X} \mid Y = j \sim \mathcal{N}(\boldsymbol{\mu}_j, \Sigma_j)$ and prior probability $p(y = j) = \pi_j$.

Illustration: x_1 is standardized humidity and x_2 is standardized pressure. Rain has higher humidity and lower pressure. Compare two covariance choices:

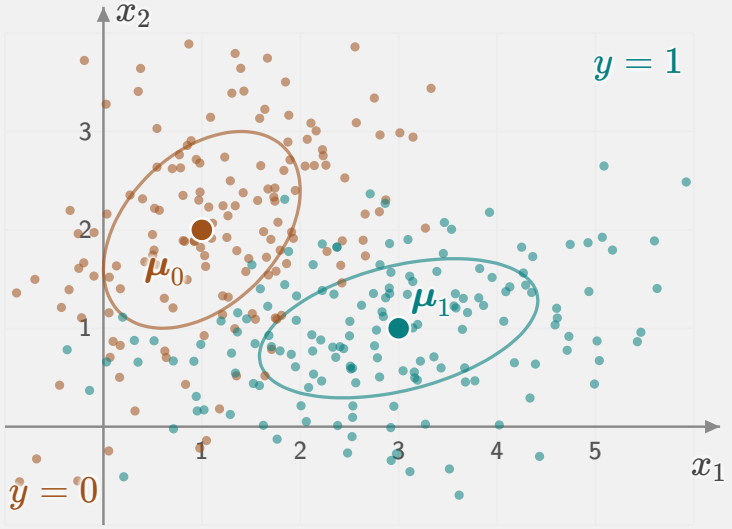
Same covariance

$$\Sigma_0 = \Sigma_1 = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 1 \end{pmatrix}$$



Different covariances

$$\Sigma_0 = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 1 \end{pmatrix}, \Sigma_1 = \begin{pmatrix} 2 & 0.4 \\ 0.4 & 0.5 \end{pmatrix}$$



LDA: Gaussian classes with a shared covariance

Suppose $\Sigma_j = \Sigma$ for every class. By Bayes' rule, with the joint density $w_j(\mathbf{x}) = \pi_j\, p(\mathbf{x} \mid y = j)$ of slide 7,

$$\ln p(y = j \mid \mathbf{x}) = \ln w_j(\mathbf{x}) - \ln p(\mathbf{x}) = \ln \pi_j + \ln p(\mathbf{x} \mid y = j) - \ln p(\mathbf{x}).$$

For the Gaussian, $\ln p(\mathbf{x} \mid y = j) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_j)^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_j)$ plus a normalizing term that depends only on Σ. Dropping that term and $-\ln p(\mathbf{x})$, which are the same for every class ($\doteq$),

$$\ln p(y = j \mid \mathbf{x}) \doteq \ln \pi_j - \tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_j)^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_j).$$

The quadratic form $(\mathbf{x} - \boldsymbol{\mu}_j)^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_j)$ is the **squared Mahalanobis distance** from $\mathbf{x}$ to $\boldsymbol{\mu}_j$ under Σ.

Expand the quadratic form:

$$\ln p(y = j \mid \mathbf{x}) \doteq -\tfrac{1}{2}\mathbf{x}^T \Sigma^{-1} \mathbf{x} + \boldsymbol{\mu}_j^T \Sigma^{-1} \mathbf{x} + \ln \pi_j - \tfrac{1}{2} \boldsymbol{\mu}_j^T \Sigma^{-1} \boldsymbol{\mu}_j.$$

The quadratic term $\mathbf{x}^T \Sigma^{-1} \mathbf{x}$ is the same for every class and cancels when log-posteriors are compared. What remains is linear in $\mathbf{x}$.

LDA: the decision boundary

For two classes, subtracting the log-posteriors of slide 10 gives

$$\ln p(y=1 \mid \mathbf{x}) - \ln p(y=0 \mid \mathbf{x}) = \boldsymbol{\beta}^T \mathbf{x} + \gamma,$$

$$\boldsymbol{\beta}^T = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \Sigma^{-1},$$

$$\gamma = \ln \frac{\pi_1}{\pi_0} - \tfrac{1}{2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \Sigma^{-1} (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_0).$$

Classification rule

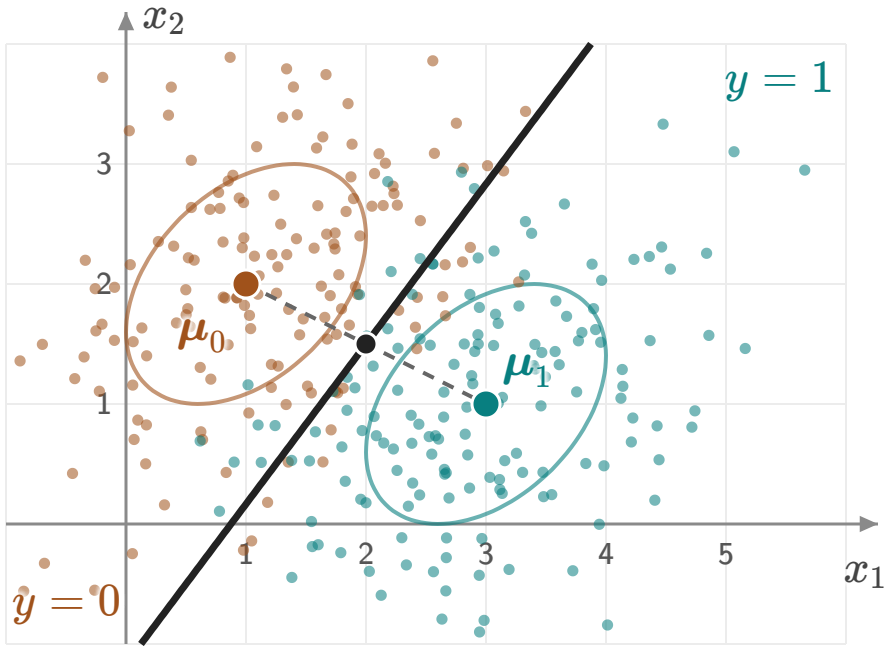
Compare $p(y=1 \mid \mathbf{x})$ and $p(y=0 \mid \mathbf{x})$:

$$\hat{y}(\mathbf{x}) = \begin{cases} 1, & \boldsymbol{\beta}^T \mathbf{x} + \gamma > 0, \\ 0, & \boldsymbol{\beta}^T \mathbf{x} + \gamma < 0. \end{cases}$$

A shared covariance makes the boundary linear

Classes tie on $\boldsymbol{\beta}^T \mathbf{x} + \gamma = 0$, a hyperplane (a line in the plane).

Figure: $\Sigma_{11} = \Sigma_{22} = 1$ and $\Sigma_{12} = \Sigma_{21} = 0.4$.



LDA's posterior is logistic

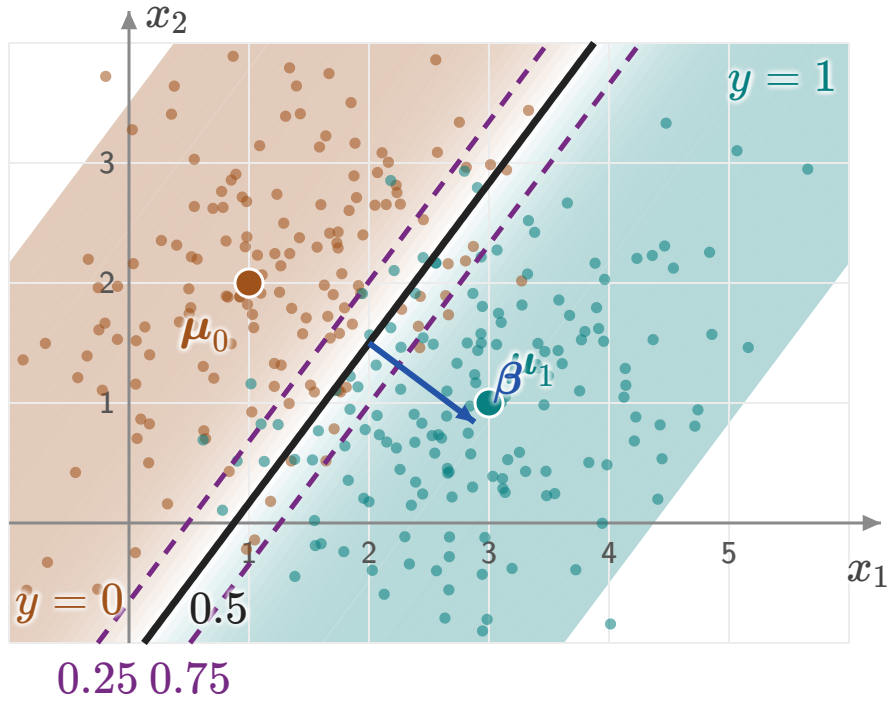
The log-posteriors differ by $\boldsymbol{\beta}^T\mathbf{x} + \gamma$, so $p(y = 0 \mid \mathbf{x})/p(y = 1 \mid \mathbf{x}) = e^{-(\boldsymbol{\beta}^T\mathbf{x}+\gamma)}$, and the two posteriors add up to 1:

$$p(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\boldsymbol{\beta}^T\mathbf{x}+\gamma)}} = \sigma(\boldsymbol{\beta}^T\mathbf{x} + \gamma).$$

Each probability level is a parallel hyperplane $\boldsymbol{\beta}^T\mathbf{x} + \gamma = \text{constant}$; the figure marks levels 0.25, 0.5 and 0.75.

Direction: $\boldsymbol{\beta}$ is normal to the boundary and points toward the fastest increase in $p(y = 1 \mid \mathbf{x})$.

Position: for fixed $\boldsymbol{\beta}$, γ shifts the boundary. Its signed distance from the origin along $\boldsymbol{\beta}$ is $-\gamma/\|\boldsymbol{\beta}\|$.



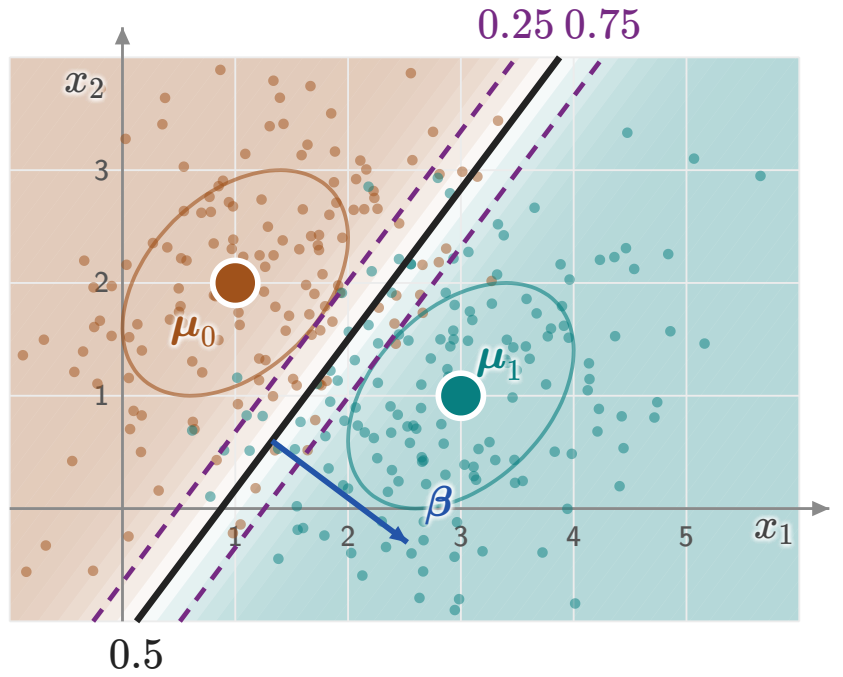
Explore the LDA posterior

Explore and explain.

1. Vary π_1: how do the joint densities and boundary change?

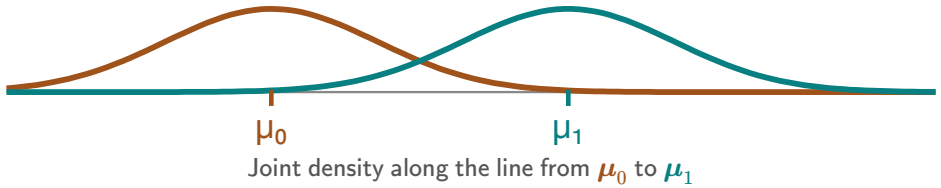
2. Drag $\boldsymbol{\mu}_0$ and $\boldsymbol{\mu}_1$: what changes?

3. At $\pi_1 = 0.5$, then 0.6, vary s and ρ in the shared covariance $\Sigma = s^2 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$. How do the boundary and posterior change?



$p(y = 0 \mid \mathbf{x}) \propto \pi_0 p(\mathbf{x} \mid y = 0)$ $\quad$ $p(y = 1 \mid \mathbf{x}) \propto \pi_1 p(\mathbf{x} \mid y = 1)$

class-conditional peak: 0.174



Joint density along the line from $\boldsymbol{\mu}_0$ to $\boldsymbol{\mu}_1$

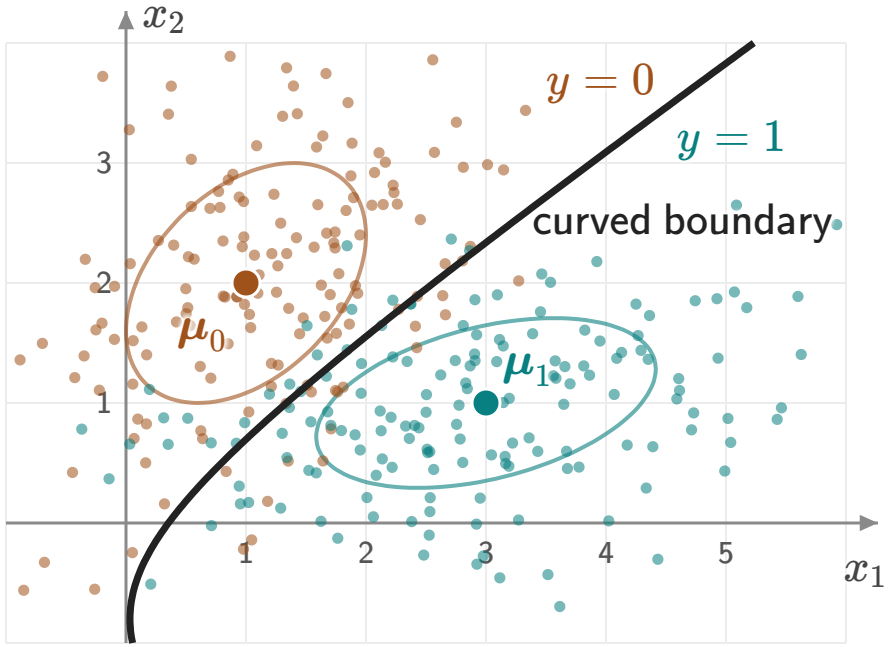
QDA, one covariance per class

With one Σ_j per class the quadratic term stays, and the boundary $\ln p(y = 1 \mid \mathbf{x}) = \ln p(y = 0 \mid \mathbf{x})$ is of the form

$$\mathbf{x}^T A \mathbf{x} + \boldsymbol{\beta}^T \mathbf{x} + \gamma = 0,$$

a curve in the plane: quadratic discriminant analysis (QDA).

Alternatively, add the quadratic features $\boldsymbol{\phi}(\mathbf{x}) = (x_1, \ldots, x_d,\ x_i x_j \text{ for } 1 \le i \le j \le d)$. A linear rule in $\boldsymbol{\phi}(\mathbf{x})$ has the same form as LDA's rule and gives a quadratic boundary in $\mathbf{x}$, without fitting a QDA density model.



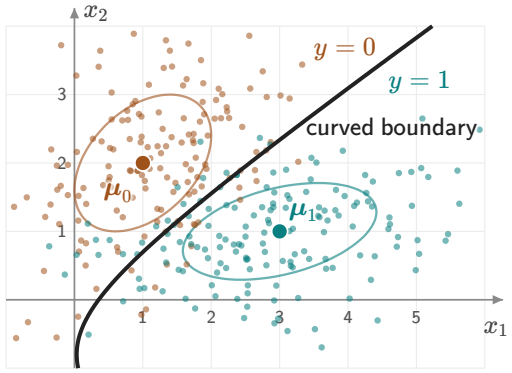
QDA, one covariance per class

With one Σ_j per class the quadratic term stays, and the boundary $\ln p(y=1 \mid \mathbf{x}) = \ln p(y=0 \mid \mathbf{x})$ is of the form

$$\mathbf{x}^T A \mathbf{x} + \boldsymbol{\beta}^T \mathbf{x} + \gamma = 0,$$

a curve in the plane: quadratic discriminant analysis (QDA).

Alternatively, add the quadratic features $\boldsymbol{\phi}(\mathbf{x}) = (x_1, \ldots, x_d,\ x_i x_j \text{ for } 1 \le i \le j \le d)$. A linear rule in $\boldsymbol{\phi}(\mathbf{x})$ has the same form as LDA's rule and gives a quadratic boundary in $\mathbf{x}$, without fitting a QDA density model.



Derivation

For general Gaussian covariances, Bayes' rule gives
$\ln p(y=j \mid \mathbf{x}) \doteq \ln \pi_j - \frac{1}{2} \ln |\Sigma_j| - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_j)^T \Sigma_j^{-1} (\mathbf{x} - \boldsymbol{\mu}_j)$.

Here $|\Sigma_1| = |\Sigma_0| = 0.84$ and $\pi_1 = \pi_0 = \frac{1}{2}$, so the log-determinant and prior terms cancel:

$$A = -\tfrac{1}{2}\left(\Sigma_1^{-1} - \Sigma_0^{-1}\right),$$
$$\boldsymbol{\beta} = \Sigma_1^{-1}\boldsymbol{\mu}_1 - \Sigma_0^{-1}\boldsymbol{\mu}_0,$$
$$\gamma = -\tfrac{1}{2}\boldsymbol{\mu}_1{}^T \Sigma_1^{-1}\boldsymbol{\mu}_1 + \tfrac{1}{2}\boldsymbol{\mu}_0{}^T \Sigma_0^{-1}\boldsymbol{\mu}_0.$$

Maximum likelihood for LDA and naive Bayes

Let n_j be the number of training examples in class j. Maximum likelihood gives the class fraction and within-class averages:

$$\hat{\pi}_j = \frac{n_j}{n}, \qquad \hat{\boldsymbol{\mu}}_j = \frac{1}{n_j} \sum_{i:y_i=j} \mathbf{x}_i.$$

Center observations at their class means. Chapter 4's Gaussian result gives the shared covariance estimate:

$$\hat{\Sigma} = \frac{1}{n} \sum_{j=0}^{1} \sum_{i:y_i=j} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_j)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_j)^T.$$

FULL LDA

Use every entry of $\hat{\Sigma}$: $d(d+1)/2$ covariance parameters.

NAIVE BAYES WITH SHARED VARIANCES

Keep only the diagonal of $\hat{\Sigma}$. Equivalently, $\hat{D}_{kk} = \frac{1}{n}\sum_i (x_{ik} - \hat{\mu}_{y_i,k})^2$: the pooled within-class sample variance of feature k (MLE; d parameters).

Full LDA vs naive Bayes

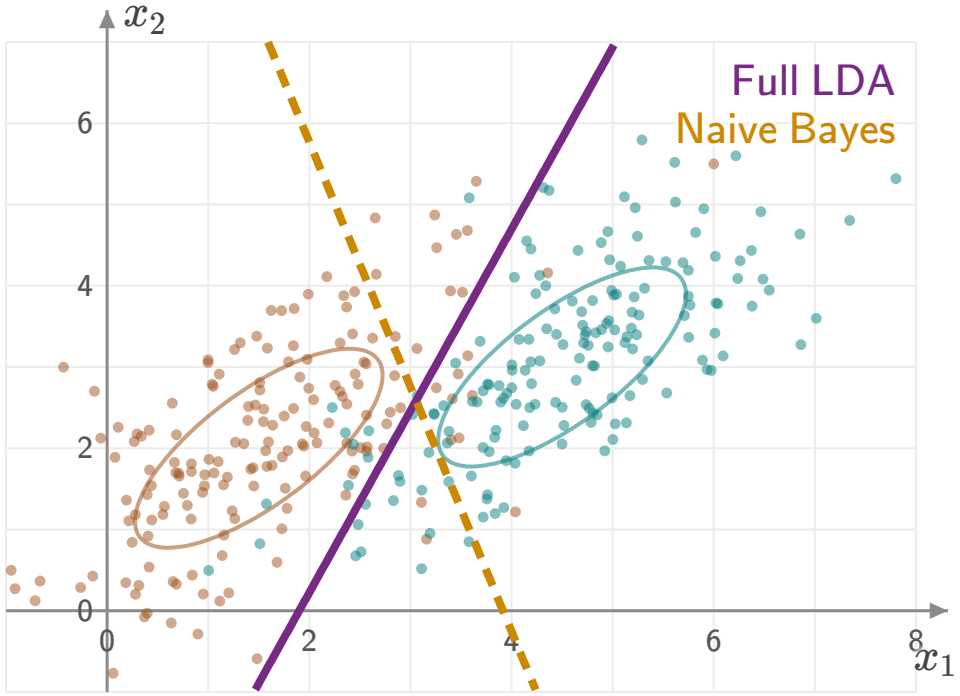
Both Gaussian classes share a covariance. Naive Bayes ignores correlations between features.

Full LDA: fit the pooled covariance $\hat{\Sigma}$.

Naive Bayes: use $\hat{D}$ with $\hat{D}_{kk} = (\hat{\Sigma})_{kk}$ and zero off-diagonal entries, so

$$p(\mathbf{x} \mid y = j) = \prod_{k=1}^{d} p(x_k \mid y = j).$$

Both boundaries are lines. LDA fits $d(d+1)/2$ covariance entries; this shared-variance naive Bayes model fits d.



Both fits use the same class means and the same training points.

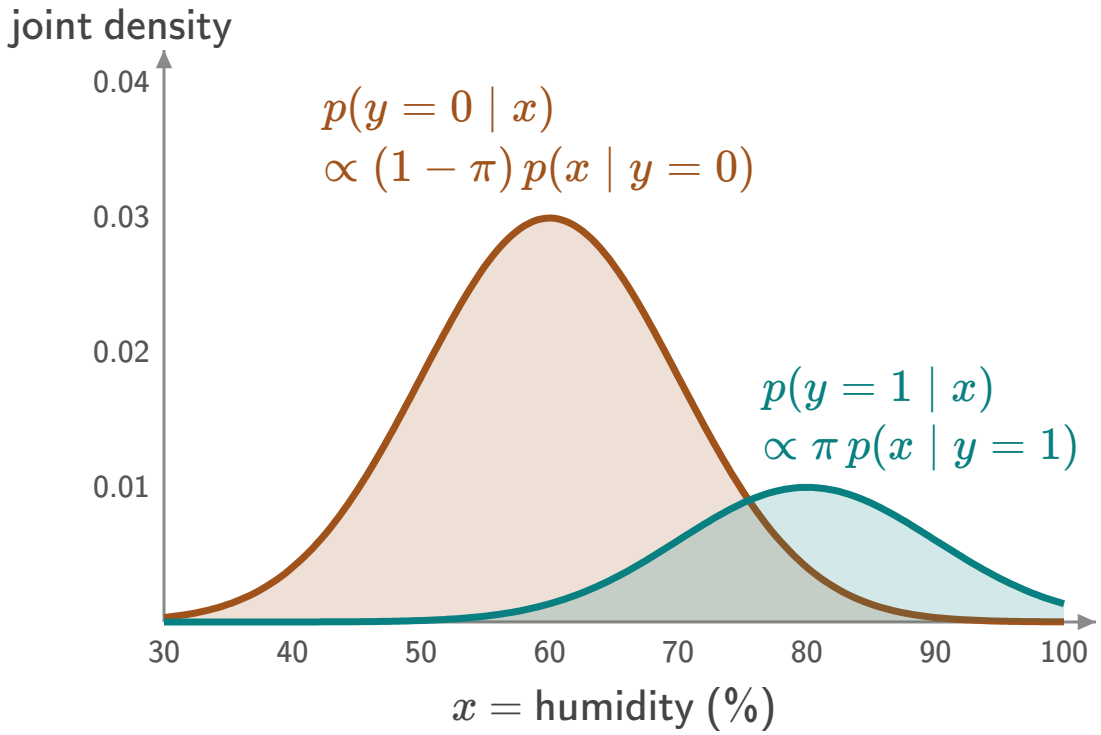
Your turn: fewer rainy days

The densities of slide 7, with $\pi = \frac{1}{4}$.

(a) What is $p(y = 1 \mid x)$ at 70% humidity, where the two class-conditional densities cross?

(b) Which way does the boundary move, and where is it?

(c) What is $p(y = 1 \mid x)$ at 80%?



Your turn: fewer rainy days (a) to (c)

Same class densities as slide 7, with rain prior $\pi = \frac{1}{4}$.

(a) Rain probability at 70%	**(b)** Decision boundary	**(c)** Rain probability at 80%

Solution

(a) The class-conditional densities are equal at $x = 70$, so the posterior equals the prior: $\frac{1}{4}$.

(b) $\ln\big(w_1(x)/w_0(x)\big) = 0.2x - 14 - \ln 3$ is zero at $x = 70 + 5\ln 3 \approx 75.49$: the boundary moves toward the rainy mean, to about 75.5%.

(c) $\sigma(0.2 \cdot 80 - 14 - \ln 3) = \sigma(2 - \ln 3) = 0.711.$

